AI HACKATHON 2026 · DIU CPC × UPAY · TRACK 05: MERCHANT & AGENT INTELLIGENCE

FloatWise

AI agent-liquidity forecaster for upay

Project report: the problem, the solution, the architecture, the technical workflow, the execution and the results.

Model version **fw-c078f4bb** · Test period 1 January – 30 June 2026 · Demo date 15 March 2026

Repository: github.com/redwanhasan980/DIU_AI_Hackathon

All data in this project is synthetic. Every number in this report is produced by code and read from `artifacts/*.json`.

Contents

1. Executive summary.... 4
Headline results.... 4
2. The problem.... 6
2.1 Context.... 6
2.2 Who loses on a failed cash-out.... 6
2.3 How big and how seasonal.... 6
2.4 Independent evidence that the problem is real.... 7
2.5 Why today's rule of thumb fails.... 7
2.6 Problem statement.... 8
3. The solution.... 9
3.1 What FloatWise does.... 9
3.2 Users.... 9
3.3 A day with FloatWise.... 9
3.4 Worked example: AG00471, five days before Eid.... 9
4. System architecture.... 11
4.1 Layers.... 11
4.2 Technology stack.... 11
4.3 Data contracts.... 12
4.4 Precomputation.... 12
5. Synthetic data.... 13
5.1 Scale and split.... 13
5.2 Generation model.... 13
5.3 Agent archetypes.... 13
5.4 Alignment with upay.... 14
5.5 Data quality checks.... 14
6. Forecasting model.... 15
6.1 Targets and models.... 15
6.2 Features (known at the forecast origin).... 15
6.3 Baselines and acceptance.... 15
7. Decision engine.... 17
7.1 Stockout probability.... 17
7.2 Recommendation.... 17
7.3 DSO allocation.... 17
7.4 Business values.... 17
8. Explainability and the language model.... 18
9. API and user interface.... 19
9.1 REST API (FastAPI, prefix /api/v1).... 19
9.2 Screens.... 19
10. Business impact.... 22
10.1 Unit economics per 1,000 agents per month.... 23
10.2 At upay's network scale.... 23
10.3 One control: the service level.... 23
10.4 Fairness.... 24
11. Design choices made on held-out data.... 25
12. Responsible AI and security.... 26
13. Execution: how it was built and how to run it.... 27
13.1 Delivery phases.... 27
13.2 Pipeline.... 27
13.3 Run it (about 6 minutes).... 27
13.4 Quality assurance.... 27
14. Limitations and next steps.... 29
14.1 What FloatWise does not fix yet.... 29
14.2 Path to a pilot.... 29
15. Validation of claims.... 30
15.1 Facts about Bangladesh, MFS and upay.... 30
15.2 Assumptions without a public source.... 31

15.3 Results....31
Appendix A. Glossary....33
Appendix B. Evidence index....34
Appendix C. References....35

1. Executive summary

Mobile financial services (MFS) agents in Bangladesh run out of **cash** (so they cannot pay cash-outs) or **e-float** (so they cannot accept cash-ins) on exactly the days when customers need them most: before Eid, on garment salary days, during remittance surges and on government allowance (G2P) days. The customer hears “cash nai, pore ashen” (“no cash, come back later”). upay loses the fee, the agent loses commission and the customer may switch wallets.

FloatWise forecasts every agent's cash-out and cash-in for the next 7 days, turns the forecast into a stockout risk and an exact top-up amount, plans the field officers' (DSOs') visits within their limits, explains each recommendation in Bangla or English, and leaves every decision to a person. A six-month backtest against today's rule of thumb measures the impact.

−47.5%
stockout agent-days

−53.7%
upay fee revenue lost

−53%
forecast error vs same day last week

Tk 9.3 Cr
upay fees recovered per month at network scale

Headline results

Result	Value	Source
Stockout agent-days, 500 agents, Jan–Jun 2026	6,973 → 3,664 (−47.5%)	simulation_default.json
Failed cash-out value	Tk 52.2 crore → Tk 27.8 crore (−46.7%)	simulation_default.json
upay fee revenue lost	Tk 33.81 lakh → Tk 15.64 lakh (−53.7%)	simulation_default.json
Cost: average cash held / DSO trips	+2.3% / +4.1%	simulation_default.json
Cash-out forecast error (WAPE), overall	18.7% vs 39.8% for same day last week	metrics.json
Cash-out forecast error on Eid and event days	21.2% vs 58.9% (−64%)	metrics.json
At upay's ~1,54,000 agents, per month	Tk 9.3 crore fee, Tk 6.4 crore commission, 1,68,938 stockout-days avoided, 2,69,962 customers kept	simulation_default.json
Fairness	Every area type and both rural and urban improve on forecast and on stockouts	fairness.json
Quality	89 automated tests, 21-step browser walk-through, CI on Linux and Windows, deterministic rebuild	repo

How to read the evidence. Facts about Bangladesh, MFS and upay are checked against independent public sources (Prothom Alo, The Daily Star, The Business Standard, The Financial Express, upay's own website, IFC/World Bank); numbers in square brackets point to the reference list in Appendix C. All results (forecast accuracy, backtest, business value) come from our own code on **synthetic** data: they are reproducible and tested, but not yet validated on real upay data. Section 15 lists every claim with its status, including one discrepancy we found.

> **Design principle.** Machine learning predicts demand. Deterministic, tested code decides amounts. The language model only explains, behind a numeric guardrail. A person approves every action; no money ever moves automatically.

2. The problem

2.1 Context

upay serves about 85 lakh customers through roughly 1,54,000 retail agents (end of 2023) [7]. When bank branches close for Eid, ATMs and MFS agents become the main way people get cash, and demand for cash withdrawals rises significantly [1]. Each agent holds a fixed total float split between physical **cash** in the drawer and **e-float** in the agent wallet. A cash-out converts the agent's cash into e-float; a cash-in does the opposite. When either side reaches zero, one type of transaction stops.

Failure mode	What happens	Typical trigger
Cash stockout	Agent cannot pay a cash-out; customer is turned away	Pre-Eid remittance arrivals, garment salary days, G2P allowance days
E-float stockout	Agent cannot accept a cash-in; sender goes elsewhere	Urban pre-Eid rush to send money home, market days

2.2 Who loses on a failed cash-out

- **The customer** cannot withdraw salary or remittance on the day it is needed and may move to another wallet.
- **The agent** loses the 0.41% commission on every failed cash-in and cash-out, on the busiest days of the year (rate from upay agent terms; not re-confirmed publicly, see Section 15).
- **upay** loses its cash-out fee (1.4% primary, 1.0% salary and remittance, 0.7% disbursement wallet [8][9]) net of the 0.51% channel commission, and risks losing the customer.
- **The distributor and DSO** lose 0.10% distributor commission and spend the day firefighting with unplanned trips.

In the six-month backtest of 500 synthetic agents, today's rule of thumb failed **2,15,101 cash-outs worth Tk 52.2 crore**.

2.3 How big and how seasonal

Under today's rule, **30%** of agent-days stock out in the week before Eid-ul-Fitr, against **1.6%** on a normal day. The pain is concentrated in a few predictable windows, which is exactly what a forecaster can exploit. These two rates come from our **synthetic** network: no public source reports agent stockout rates, so their size is an assumption, while the direction (demand concentrated around Eid, salary and allowance days) is supported by the sources in Section 2.4.

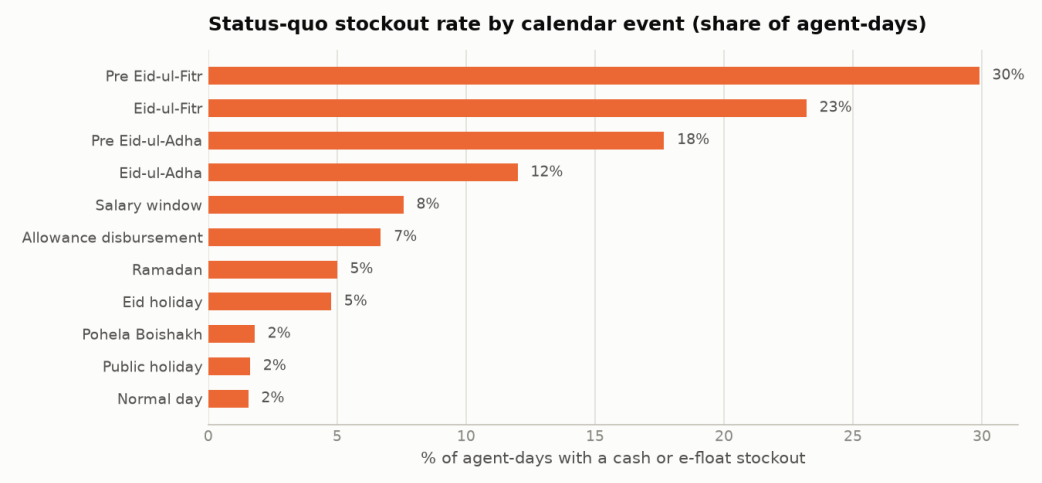


Figure 1. *Stockout rate under today's rule by type of day (synthetic network, full dataset).*

2.4 Independent evidence that the problem is real

Fact	Evidence	Source
Cash demand rises sharply at Eid; agents and ATMs are the main cash access when banks close	“Demand for cash withdrawals from ATM booths rises significantly during Eid… ATM booths and MFS cash-out facilities become the primary means of accessing money.”	Prothom Alo, 16 Mar 2026 [1]
Cash demand surges before Eid-ul-Adha; companies withdraw cash for bonuses	“Demand for cash rises sharply around cattle markets during Eid-ul-Adha. Companies also withdraw cash to pay bonuses…”	The Business Standard, 16 May 2026 [5]
Remittances peak before Eid	Record USD 2.94 billion in the first 26 days of March 2025; “Usually, remittances peak before Eid.”	Prothom Alo, 27 Mar 2025 [2]
Record monthly remittance driven by Ramadan and Eid	USD 3.29 billion in March 2025, “due to expatriates sending more money… marking… Eid-ul-Fitr”	The Business Standard, 6 Apr 2025 [3]
MFS volumes peak in Eid months; cash-in/cash-out dominate	Record Tk 1.25 lakh crore in April 2023 (Ramadan and Eid); cash-in and cash-out were 59% of transactions	The Daily Star, 2023 [4]
Salaries reach workers through MFS	About 11 lakh garment workers were paid digitally in 2020; about 40% of workers are paid digitally	The Business Standard, 16 Mar 2021 [6]
Government allowances (G2P) are paid into MFS wallets	Over 75 lakh people received about Tk 1,624 crore of allowances through one MFS provider in FY26	The Business Standard, 5 Nov 2025 [10]
Agent liquidity is a core problem of mobile money worldwide	“Solving the liquidity management challenge is one of the next big issues facing mobile money”; liquidity is the agents' primary cost (CGAP research)	IFC Mobile Money Toolkit [11]

2.5 Why today's rule of thumb fails

Agents and DSOs top up to about 1.5× the last 7 days' average when the balance falls below 0.6× of that target. An average only sees the past. Eid moves 10–11 days earlier every year (11 April 2024, 31 March 2025, 21 March 2026 in Bangladesh [12]), and salary and allowance days

follow the calendar, not the average. For the demo agent the average keeps climbing during the pre-Eid rush and peaks on Eid day, after demand has already fallen by about three-quarters.

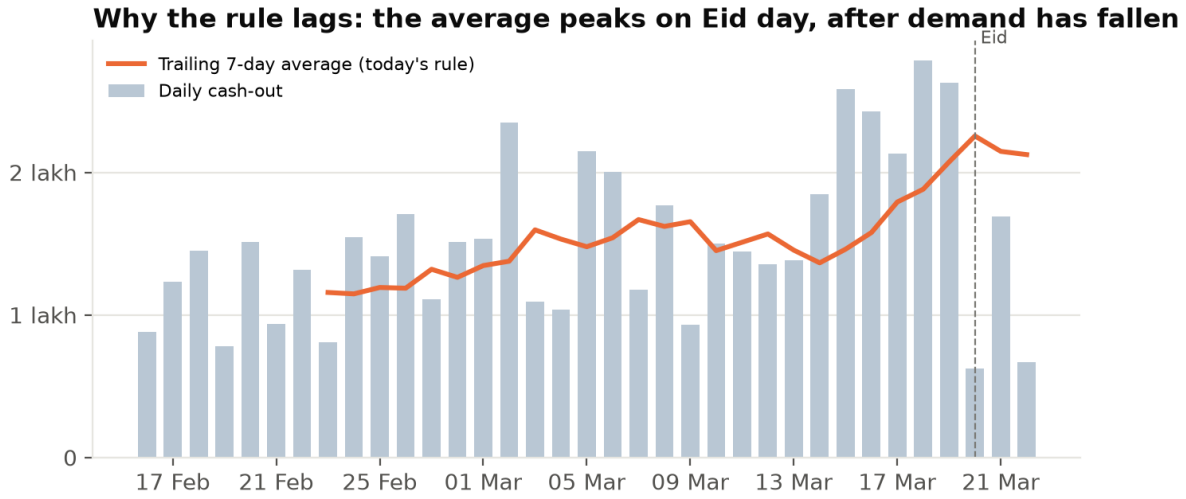


Figure 2. *Demo agent AG00471: daily cash-out against the trailing 7-day average used by today's rule.*

2.6 Problem statement

> **For** upay operations, distributors, DSOs and agents, **who** cannot see tomorrow's cash and e-float needs, FloatWise **is** an AI liquidity forecaster **that** predicts each agent's 1–7-day demand, sizes the top-up and plans DSO visits, **unlike** a backward-looking 7-day average, **because** it learns the Bangladesh calendar (Eid, salary, remittance, allowance days) and reports its uncertainty.

3. The solution

3.1 What FloatWise does

1. **Forecast.** Every agent's cash-out and cash-in for the next 1–7 days as quantiles (P10, P50, P90).
2. **Assess risk.** Convert the forecast into the chance of running out of cash or e-float tomorrow and over the next 7 days.
3. **Recommend.** TOP_UP_CASH, TOP_UP_EFLOAT, NO_ACTION or NEEDS_EXTRA_FLOAT, with an exact amount rounded up to Tk 1,000 and reason codes.
4. **Plan.** Rank DSO visits by taka at risk within the visit limit (10) and the cash bag (Tk 10 lakh); unvisited agents get the advice in the app.
5. **Explain and approve.** SHAP drivers plus a Bangla or English explanation; a person approves, modifies or rejects.
6. **Prove.** Backtest FloatWise against today's rule on six months of true demand, including Eid-ul-Fitr and Eid-ul-Adha 2026.

3.2 Users

Persona	Needs	FloatWise screen
Operations lead (upay / distributor)	Where is the network at risk, how much cash and e-float to move, is it working?	Network map, Impact, Model & fairness
DSO (field officer)	Which agents to visit today with a limited bag	DSO plan
Agent	How much cash to keep tomorrow and why, in Bangla	Agent mobile card
Risk / compliance	Is it fair, explainable and human-controlled?	Model & fairness, Decisions log

3.3 A day with FloatWise



Figure 3. *The daily operating loop.*

3.4 Worked example: AG00471, five days before Eid

Rahmania Bhandar (AG00471) is a rural remittance shop in Beanibazar, Sylhet. At close on 15 March 2026 it holds **Tk 18,501** in cash. FloatWise forecasts **Tk 2,60,567** of cash-out for 16 March (P10 Tk 2,14,385 – P90 Tk 3,03,469), a 100% chance of running out. It recommends **TOP_UP_CASH Tk 2,66,000 before opening**, limited by the agent's total float of Tk 2,88,000. The main SHAP drivers are days to Eid (+Tk 95,410) and Ramadan (+Tk 13,725). True demand fell inside the P10–P90 range on 6 of the 7 forecast days.

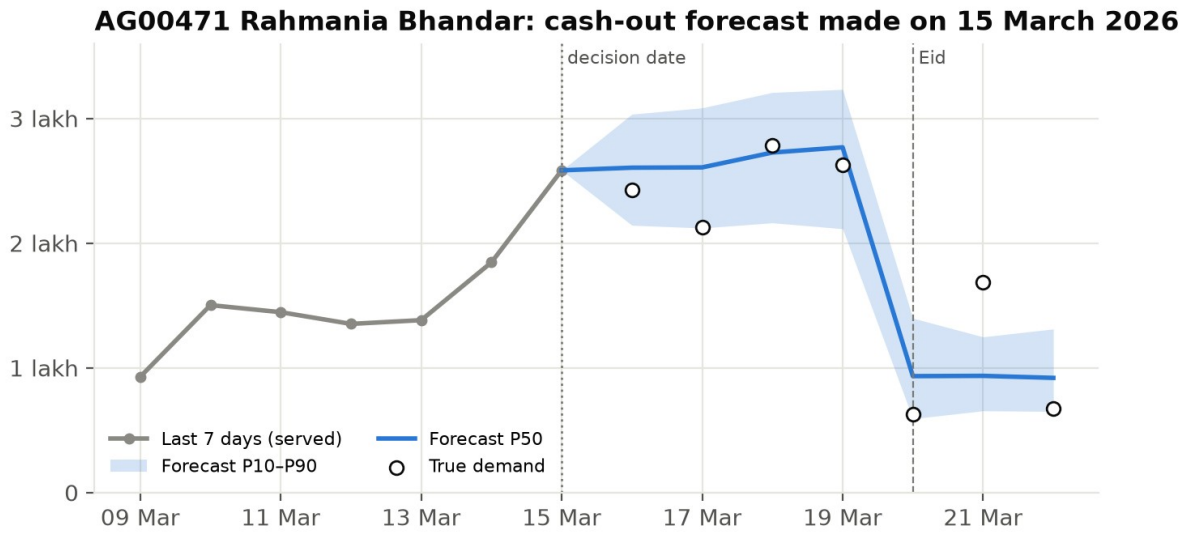


Figure 4. *Forecast made on 15 March against what really happened.*

4. System architecture

FloatWise is a layered Python backend with a React frontend. Lower layers never import higher ones, and all business rules live in one place (`engine/`).

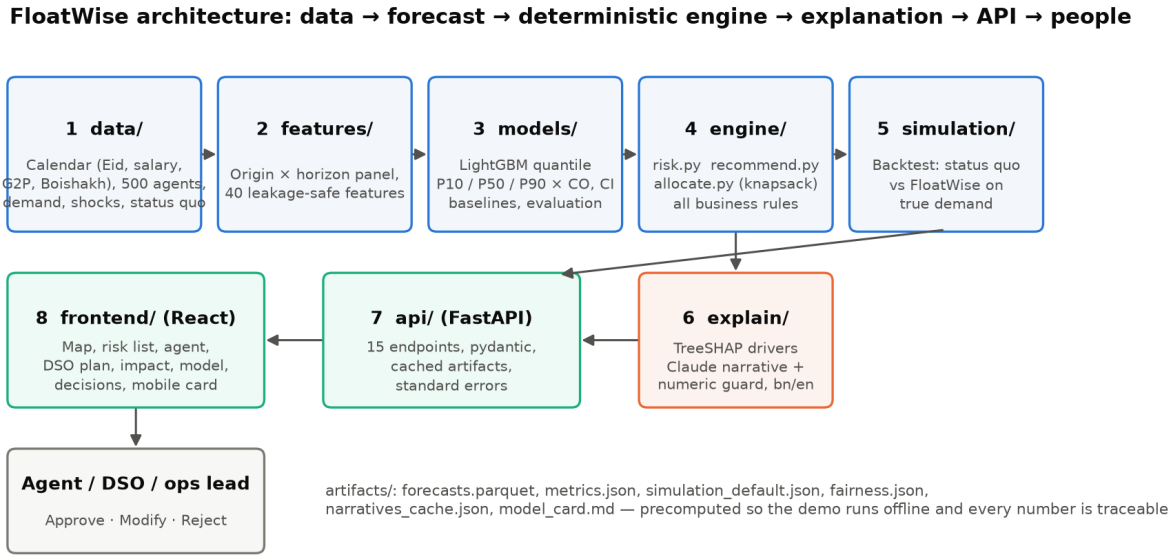


Figure 5. *Layers and data flow.*

4.1 Layers

Layer	Responsibility	Key modules
data/	Bangladesh calendar, 500 agents, true demand, shocks, censored ledger under today's rule	calendar_bd.py, agents.py, demand.py, shocks.py, simulate_status_quo.py, generate.py
features/	Origin × horizon panel with 40 leakage-safe features	build.py
models/	Baselines, LightGBM quantile training, evaluation, batch prediction	baselines.py, train.py, evaluate.py, predict.py
engine/	Stockout probability, recommendation, DSO allocation (all business rules)	risk.py, recommend.py, allocate.py
simulation/	Day-by-day backtest of status quo vs FloatWise on true demand	backtest.py
explain/	TreeSHAP drivers, Bangla/English templates, Claude narrative with guardrail	shap_explain.py, templates.py, llm.py, labels.py
api/	FastAPI service over precomputed artifacts; also serves the built web app	main.py, store.py
frontend/	React + TypeScript single-page app with offline fixtures	pages/, components/, api/, i18n/, fixtures/

4.2 Technology stack

Area	Technology
Language and runtime	Python 3.11+, Node.js 20+
Data and ML	pandas 3.0, NumPy 2.4, PyArrow (Parquet), LightGBM 4.7 (quantile

Area	Technology
	objective), scikit-learn 1.9
Explainability and LLM	LightGBM TreeSHAP (pred_contrib); Anthropic Claude SDK (model from LLM_MODEL env var)
API	FastAPI 0.142, pydantic 2.13, Uvicorn
Frontend	React 19, TypeScript 5.9 (strict), Vite 8, Tailwind CSS 4, Recharts 3, react-leaflet 5 with an offline Bangladesh outline
Quality	pytest (89 tests), ruff, tsc, eslint, Playwright browser walk-through, GitHub Actions (Linux + Windows)
Tooling	run.py cross-platform launcher, Makefile, config.yaml for every assumption

4.3 Data contracts

The pydantic schemas in `backend/floatwise/schemas.py` are mirrored one-to-one in `frontend/src/api/types.ts`. Any contract change updates the SRS, both schema files and the frontend fixtures in the same commit. Money is always an integer number of taka.

4.4 Precomputation

Forecasts for the test period are generated once into `artifacts/forecasts.parquet`. The API reads them, so every GET responds in under 300 ms and the demo runs without internet. The artifacts are deterministic: rebuilding from the same seed reproduces model `fw-c078f4bb` and every number byte for byte.

5. Synthetic data

No real customer, agent or transaction data is used. The generator is deterministic from a seed in `config.yaml`, and every assumption is documented in `docs/DATA_ASSUMPTIONS.md`.

5.1 Scale and split

Item	Value
Period	1 January 2024 → 30 June 2026 (daily)
Agents	500 across 15 districts, about 25 per DSO, 5 DSOs per distributor
Train / validation / test	≤ 30 June 2025 / 1 July – 31 December 2025 / 1 January – 30 June 2026
Eid-ul-Fitr	11 April 2024, 31 March 2025, 20 March 2026 (Bangladesh celebrated on 21 March 2026 [12]; see Section 15)
Eid-ul-Adha	17 June 2024, 7 June 2025, 27 May 2026 (Bangladesh celebrated on 28 May 2026 [13]; see Section 15)

5.2 Generation model

True daily demand is a product of interpretable multipliers:

> `demand_CO(agent, d) = base[archetype] × size[agent] × weekday × salary(day index) × eid(days to/after Eid) × ramadan × boishakh × g2p × shock(district, d) × trend(d) × lognormal noise`
>
> Cash-in follows the same pattern with its own multipliers. The ledger then plays today's rule of thumb against this demand; on stockout days the observed volume is **censored** (lower than true demand), just as in real data.

5.3 Agent archetypes

Archetype	Base cash-out / day	Base cash-in / day	Signature
urban_residential (125)	Tk 1,50,000	Tk 1,80,000	Cash-in spike before Eid → e-float stockouts
garment_belt (100)	Tk 2,50,000	Tk 1,00,000	Cash-out spike on salary days 1–10 and the Eid bonus
rural_remittance (150)	Tk 80,000	Tk 40,000	Cash-out ×2–3 before Eid as remittance arrives; G2P days
market_hub (75)	Tk 3,00,000	Tk 3,00,000	High volume both ways; Boishakh and pre-Eid shopping
university_area (50)	Tk 60,000	Tk 50,000	Month-start cash-in from families; vacations

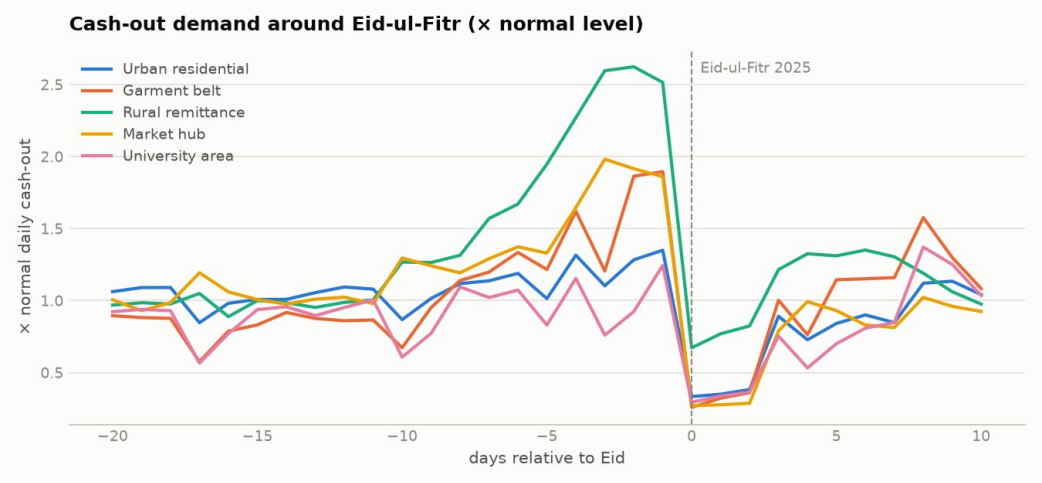


Figure 6. *Eid spike by area type in the generated data.*

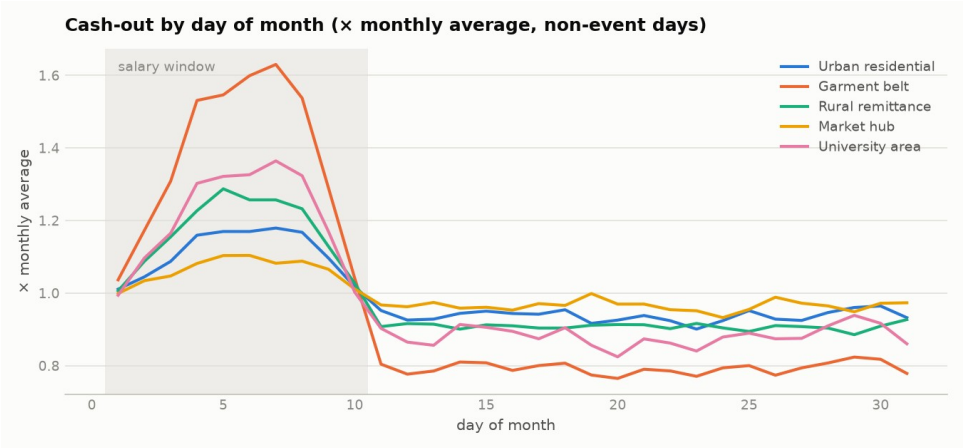


Figure 7. *Salary-day cycle for garment-belt agents.*

5.4 Alignment with upay

upay reality (public)	In FloatWise
Agent cash-out fee by wallet: primary 1.4%, salary 1.0%, remittance 1.0%, disbursement 0.7%	Fee per area type by wallet mix (garment → salary wallet, rural → remittance/G2P blend)
Agent commission 0.41%, distributor 0.10% per cash-in / cash-out	Commission lost on failed transactions; upay keeps fee − 0.51%
Salary disbursement, inward remittance, G2P allowances	Garment salary days, pre-Eid remittance surge, quarterly allowance days
Distributor → DSO → agent network, ~1,54,000 agents	500 agents, 15 districts; impact scaled to 1,54,000
Customer cash-out limit Tk 25,000 per transaction	Synthetic ticket sizes Tk 1,500–3,000

5.5 Data quality checks

`tests/test_data_quality.py` checks that the same seed gives identical files, that cash + e-float equals the agent's float every day, that no balance goes negative, that served volume never exceeds true demand, that stockouts concentrate on event days, and that the Eid, salary and G2P signatures appear where expected.

6. Forecasting model

6.1 Targets and models

- Two targets (cash-out, cash-in) × three quantiles (P10, P50, P90) = six global **LightGBM** models with the quantile objective.
- Direct multi-horizon: the horizon h = 1..7 is a feature, so one model covers the whole week.
- The target is the ratio of demand to the agent's trailing 28-day mean, so one model learns multipliers across agents of very different size.
- Crossing quantiles are re-ordered (rearrangement) so that P10 ≤ P50 ≤ P90 always holds.

6.2 Features (known at the forecast origin)

Group	Features
Horizon	h
Target-date calendar	weekday, Friday, days to Eid-ul-Fitr / Eid-ul-Adha, days after Eid, Ramadan, salary-day index and window, Boishakh, holiday, G2P day index
Lags (≤ origin)	last value, lags 1/6/13, rolling mean/std 7/14/28, rolling max 28
Event memory	agent's multiplier around the previous Eid
Agent static	archetype, float capacity, district, DSO density
Censoring	stockout count in the last 7 days

No leakage: a feature for origin t may use data up to t only; calendar features of the target date are allowed because they are known in advance. `tests/test_no_leakage.py` enforces this by perturbing future data and checking that features do not change.

6.3 Baselines and acceptance

Every model is compared with two baselines: **same day last week** (seasonal naive) and the **trailing 7-day mean** (today's rule). All four acceptance checks pass.

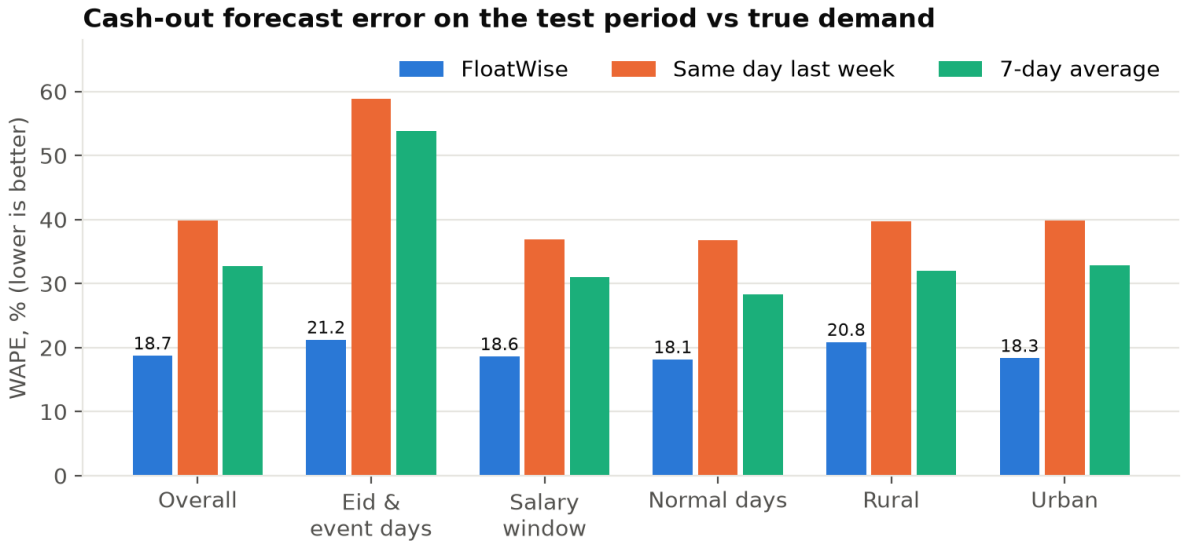


Figure 8. *Cash-out WAPE on the test period, measured against true (uncensored) demand.*

Acceptance check (SRS §6.4)	Target	Result
Overall improvement vs same day last week	≥ 20%	53% (18.7% vs 39.8%)
Event-day improvement	≥ 35%	64% (21.2% vs 58.9%)
P10–P90 coverage	72–88%	75% overall (60% on event days, reported as a limit)
No archetype worse than the baseline	all five	all five better

7. Decision engine

The engine is deterministic, unit-tested code. It turns quantile forecasts and balances into actions and amounts.

7.1 Stockout probability

A piecewise-linear distribution is built from P10, P50 and P90 with explicit tails. The effective opening cash is `open_cash + α × CI_P50` (α = 0.5 for same-day inflows). Its position in the cash-out distribution gives `p_cash_stockout`; the e-float side is symmetric. Risk bands: high ≥ 0.5, medium ≥ θ, else low.

7.2 Recommendation

> `target_cash(s) = ceil_1000( CO_q(s) − α·CI_P50 + Tk 5,000 buffer )`, and symmetrically for e-float.
>
> If cash and e-float targets fit within the agent's total float: top up the risky side to its target.
>
> If they do not fit: apply the best pro-rata split of the float.
>
> If even the P90-level needs exceed the float: **NEEDS_EXTRA_FLOAT**, always escalated to a person.
>
> Otherwise: NO_ACTION. Defaults: service level s = 0.99 and risk threshold θ = 0.10, chosen on validation.

Each recommendation carries reason codes from the calendar and balances (for example `EID_FITR_IN_5_DAYS`, `RAMADAN`, `SALARY_WINDOW`, `CASH_BELOW_NEED`, `RECENT_STOCKOUTS`) and a 7-day outlook of risk and need.

7.3 DSO allocation

Candidates are a DSO's agents with a top-up action. Priority is the **expected shortfall in taka**, computed exactly from the forecast distribution. An exact knapsack optimiser picks the visits that cover the most taka at risk within 10 visits and a Tk 10 lakh cash bag; a greedy planner is kept for comparison. Agents not visited receive the advice in the app, and the backtest assumes 70% of them follow it.

7.4 Business values

Value	Setting
Cash-out fee	1.0–1.4% by area type (upay published tariffs by wallet)
Channel commission	agent 0.41% + distributor 0.10% = 0.51%
Customer churn after a failed cash-out	5% (assumption, labelled in the UI)
Opportunity cost of idle cash	10% a year (assumption)
Network scale-up	~1,54,000 upay agents

8. Explainability and the language model

- **Drivers:** LightGBM TreeSHAP contributions of the median model, converted to taka and shown as the top five drivers with plain labels in both languages.
- **Narrative:** a deterministic Bangla or English template always exists. With an API key, Claude writes a friendlier explanation from structured JSON facts only (no free text from users).
- **Numeric guardrail:** every number in the LLM's text must appear in the facts; otherwise, on refusal, timeout or error, the template is used. The explanation endpoint never fails.
- **Caching:** narratives for the demo are pre-generated into `artifacts/narratives_cache.json`, so the demo works offline.
- **Labelling:** the UI marks the text as AI-generated or template, and keeps predictions, assumptions and generated text visually distinct.

> **The LLM never decides.** Amounts, risk and actions come only from `engine/`. The model explains them in the agent's language.

9. API and user interface

9.1 REST API (FastAPI, prefix /api/v1)

Method	Endpoint	Purpose
GET	/health	Status, model version, data range, demo date, LLM availability
GET	/agents, /agents/{id}	Agent list with risk and action; profile, balances and 28-day history
GET	/agents/{id}/forecast	7-day cash-out / cash-in quantiles with actuals when available
GET	/agents/{id}/recommendation	Action, amount, probabilities, reason codes, 7-day outlook
GET	/agents/{id}/explanation	SHAP drivers and Bangla/English narrative
GET	/dsos, /dso/{id}/plan	DSO list; visit plan under max_visits and max_cash
GET	/network/summary	Risk bands, actions, districts, upcoming events
POST	/simulate	Backtest with custom policy parameters
GET	/metrics/model, /calendar	Evaluation, fairness, tuning, assumptions; events
POST / GET	/decisions, /decisions/summary	Approve / modify / reject log and feedback summary

Inputs are validated with pydantic, errors use one standard JSON format without stack traces, and CORS is restricted to the frontend origin.

9.2 Screens

Screen	What it shows
Network	Eid banner, KPI tiles, risk map of 500 agents, districts by risk
Risk list	Filterable list of agents by district, area type and risk
Agent detail	Recommendation, 7-day outlook, forecast vs true demand, drivers, explanation, approve/modify/reject
DSO plan	Visits ranked by taka at risk within the bag; excluded agents and why
Impact	Backtest, upay fee revenue, network scale-up, live re-run with the service-level slider
Model & fairness	Acceptance checks, error by slice, fairness table, policy tuning, assumptions
Decisions	Decision log and feedback summary
Agent mobile card	One clear Bangla action for the agent

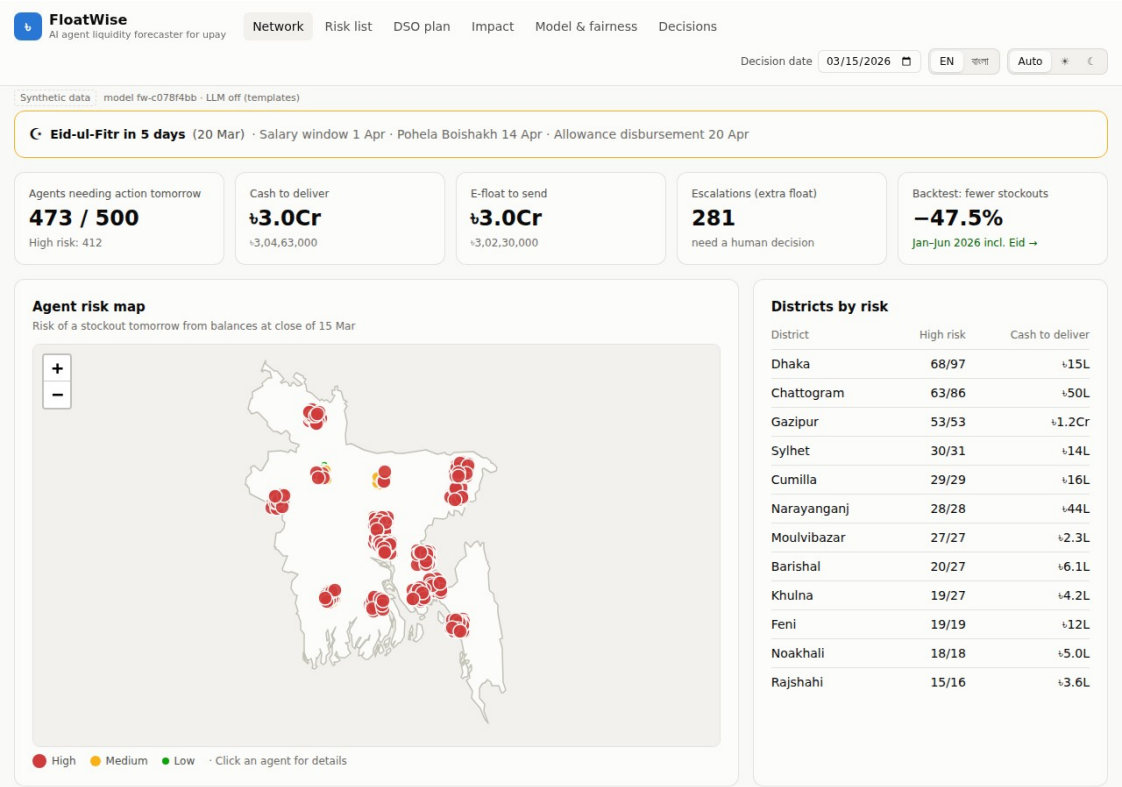


Figure 9. *Network screen on 15 March 2026.*

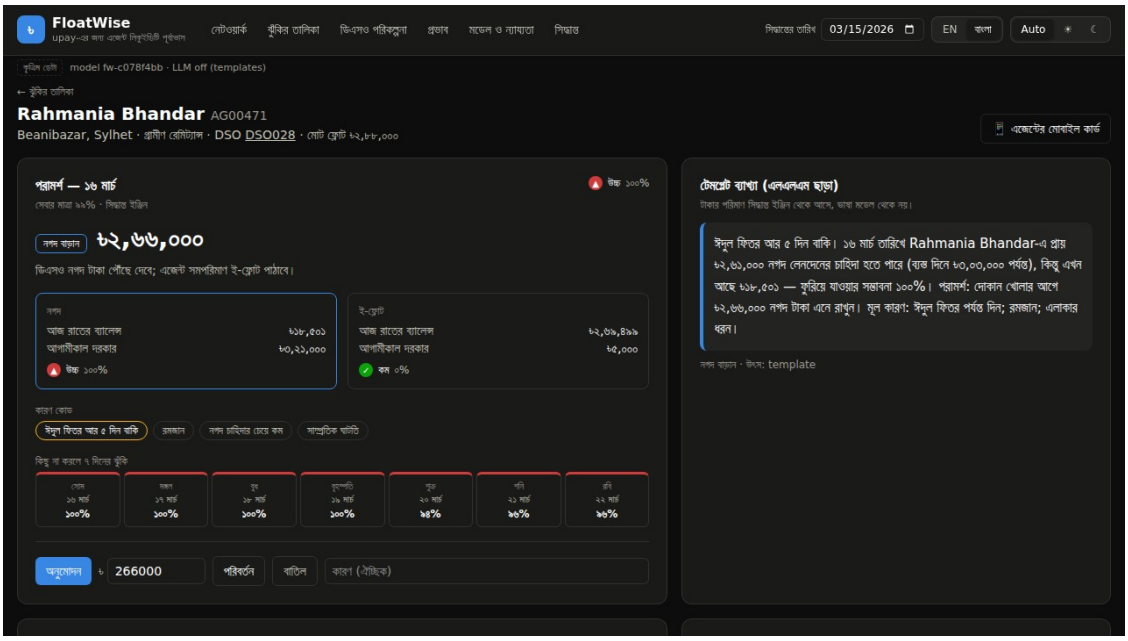


Figure 10. *Agent detail in Bangla (dark mode).*

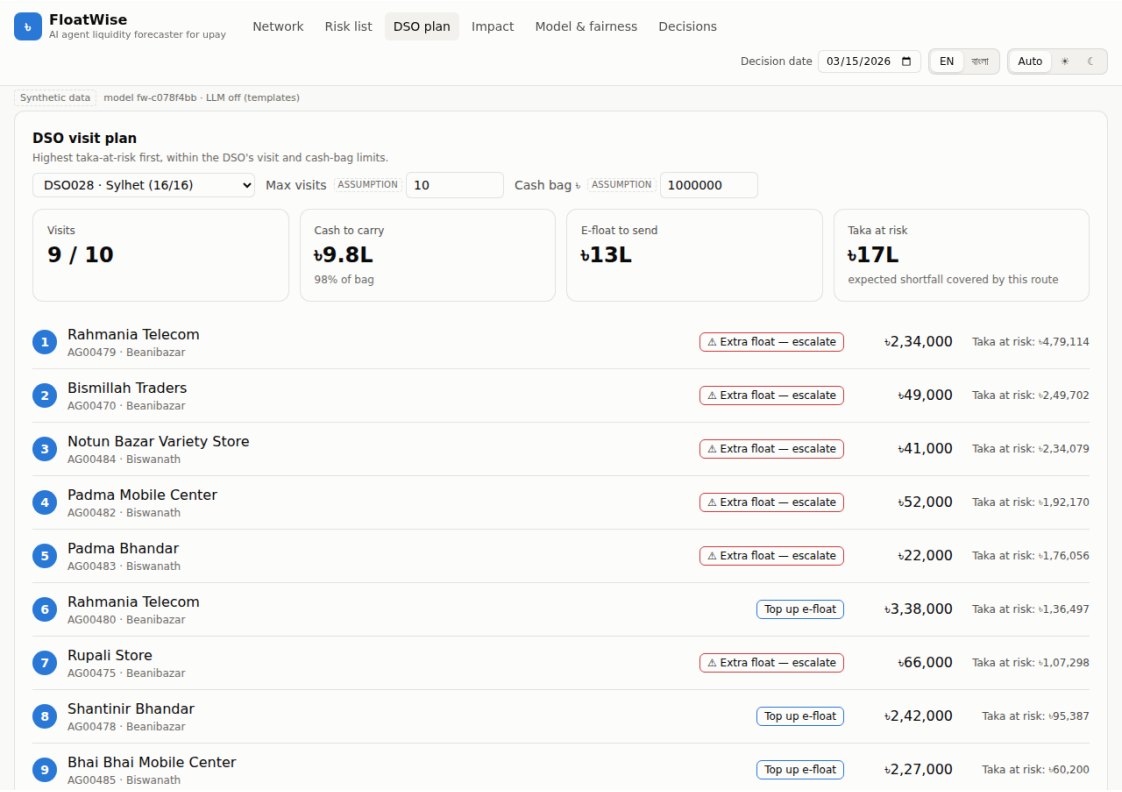


Figure 11. *DSO plan within the visit and cash-bag limits.*

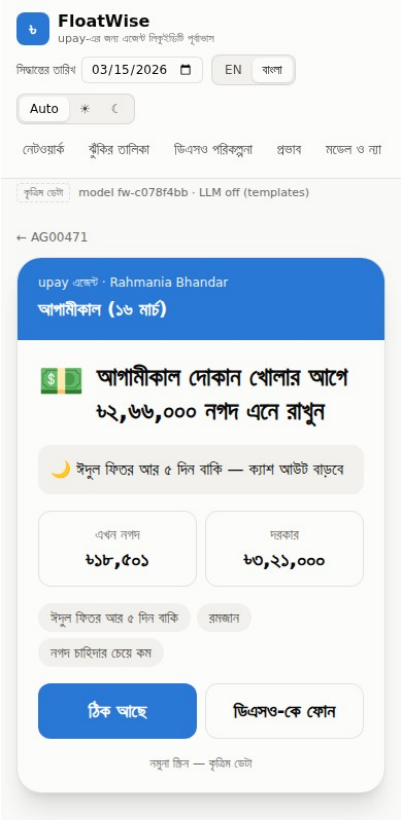


Figure 12. *Agent mobile card in Bangla.*

10. Business impact

The backtest replays January–June 2026 day by day on true demand. Both policies start from the same balances and float; only the timing and size of each top-up differ. The test period was never used for tuning.

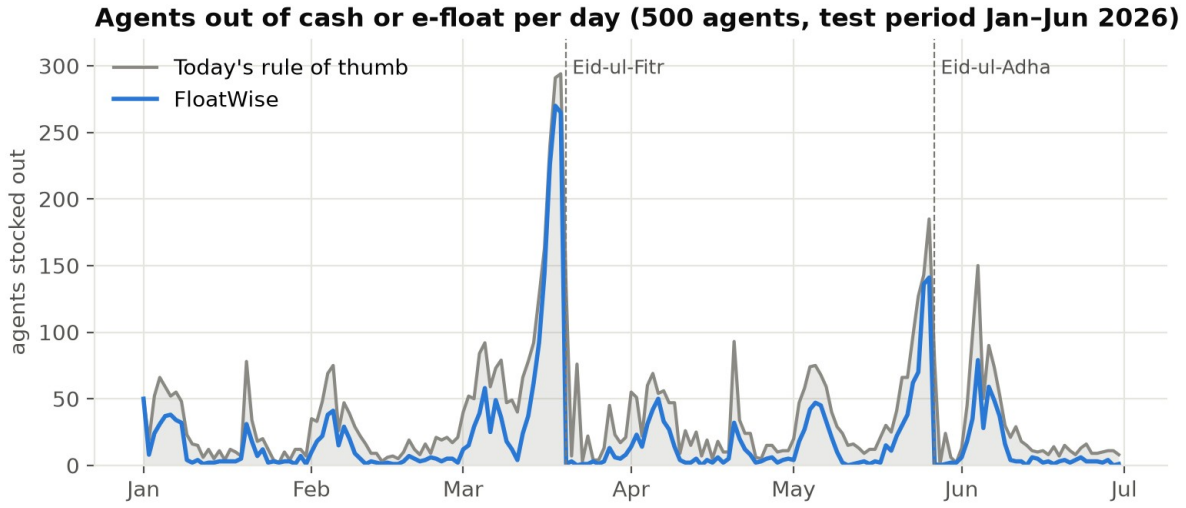


Figure 13. *Agents out of cash or e-float per day.*

Metric (500 agents, 6 months)	Today's rule	FloatWise	Change
Stockout agent-days	6,973	3,664	-47.5%
Failed cash-outs (count)	2,15,101	1,09,349	-49.2%
Failed cash-out value	Tk 52.2 crore	Tk 27.8 crore	-46.7%
upay fee revenue lost	Tk 33.81 lakh	Tk 15.64 lakh	-53.7%
Agent + distributor commission lost	Tk 27.13 lakh	Tk 14.50 lakh	-46.6%
Customers at risk of leaving	10,755	5,467	-49.2%
Share of demand served	98.7%	99.3%	+0.59 pts
Average cash held per agent	Tk 3.53 lakh	Tk 3.61 lakh	+2.3%
DSO rebalancing trips	49,813	51,860	+4.1%

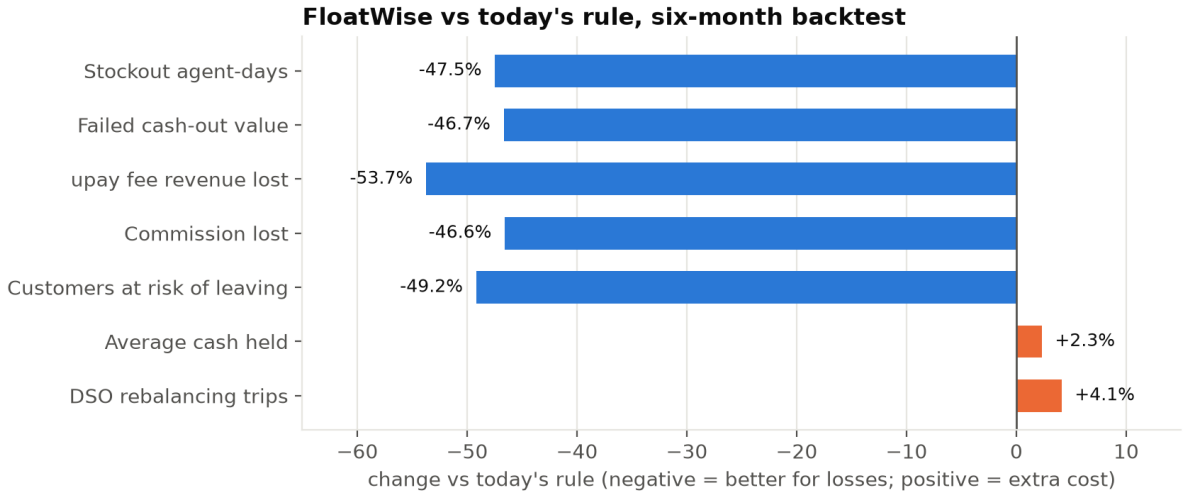


Figure 14. *Change against today's rule: losses fall by about half for a small cost in cash and trips.*

10.1 Unit economics per 1,000 agents per month

Gains	Value	Costs	Value
Stockout-days avoided	1,097	Extra DSO trips	679
Failed transactions avoided	35,742	Extra idle-cash cost (10%/yr)	Tk 67,867
Cash-outs served instead of refused	Tk 8.2 crore		
upay fee revenue recovered	Tk 6.02 lakh		
Commission for agents and distributors	Tk 4.19 lakh		
Customers kept	1,753		

10.2 At upay's network scale

Tk 9.3 Cr
upay fees / month

Tk 6.4 Cr
commission / month

1.69 lakh
stockout-days avoided / month

2.7 lakh
customers kept / month

Per-1,000-agent results scaled to ~1,54,000 agents. Customers kept assumes 5% churn after a failed cash-out.

10.3 One control: the service level

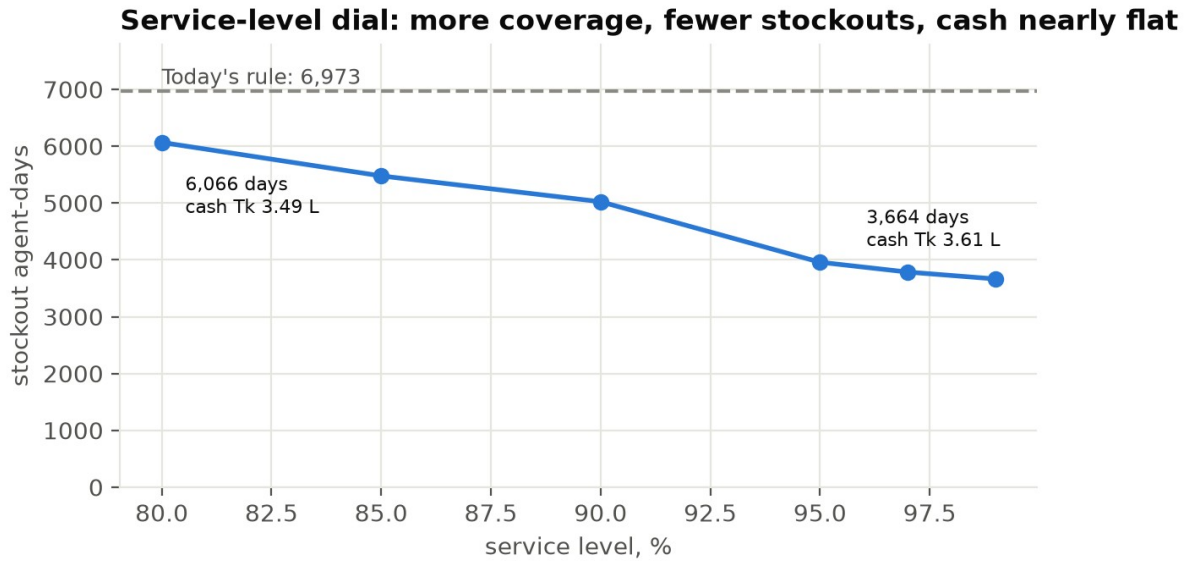


Figure 15. *Service-level sweep (shown on the test period for illustration; 0.99 was chosen on validation).*

10.4 Fairness

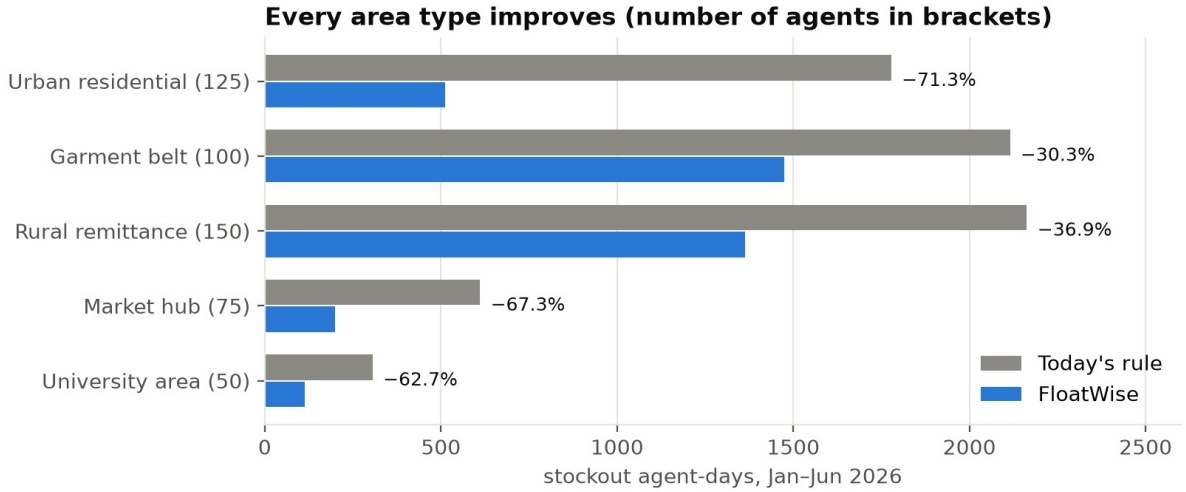


Figure 16. *Stockout agent-days by area type.*

Every segment improves. Rural remittance and garment-belt agents gain least because their total float is small relative to the Eid and salary peaks; those cases are escalated as NEEDS_EXTRA_FLOAT rather than hidden.

11. Design choices made on held-out data

No choice was made on the test period. Each was decided by a script with a written rule, and its evidence is committed.

Choice	Options	Decided on	Selected	Evidence
Stockout-day labels (censoring)	keep / exclude / inflate	Held-out Eid 2025	keep: fewest pre-Eid misses above P90 (57.1% vs 57.4% / 59.5%)	censoring_experiment.json
Quantile architecture	independent / spread	Held-out Eid 2025	independent: spread cut Eid coverage 51% → 42%	quantile_experiment.json
Service level & risk threshold	15-point grid	Validation Jul–Dec 2025	0.99 / 0.10: −55% stockouts, +12.4% trips, +2.4% cash	policy_tuning.json
DSO allocator	greedy / exact knapsack	Validation Jul–Dec 2025	optimal (1,689 vs 1,692 stockout-days)	allocator_comparison.json

12. Responsible AI and security

Principle	How FloatWise implements it
Synthetic data only	No personal data, real names, phone numbers or NIDs anywhere; generic shop names
LLM explains, never decides	Amounts from engine/; LLM gets structured JSON only; numeric post-check with template fallback
Human in the loop	Approve / Modify / Reject on every action; NEEDS_EXTRA_FLOAT always escalates; no automatic money movement
Fairness	Forecast and service checked per area type and rural vs urban; worse segments must be reported
Transparency	AI text labelled; predictions, assumptions and generated text visually distinct; model card published
Honest evaluation	Time-based split; baselines; test period untouched until final evaluation; missed targets reported
Scope	Operational liquidity only; no credit, lending or penalty decisions about agents
Security	Secrets only in .env; pydantic validation; standard errors without stack traces; restricted CORS; works offline

13. Execution: how it was built and how to run it

13.1 Delivery phases

Phase	Delivered
0–1 Contracts	SRS, implementation plan, rules for AI agents, config, schemas, Makefile
2 Data	Calendar, agents, demand, shocks, status-quo ledger; data-quality tests; check plots
3–4 Models	Feature panel with leakage test; baselines; LightGBM quantile models; evaluation and model card
5–6 Engine and simulation	Risk, recommendation, DSO allocation; backtest; policy tuning on validation
7–8 Explain and API	SHAP drivers, templates, Claude narrative with guardrail; FastAPI with 15 endpoints
9 Frontend	Eight screens, Bangla/English, light/dark, offline fixtures
10 Fairness and feedback	Fairness report; decision log and feedback summary
11 Hardening	Offline demo, screenshots, CI on Linux and Windows, fresh-clone rebuild, e2e walk-through, upay alignment

13.2 Pipeline

1. `floatwise.data.generate`: synthetic data into data/generated/.
2. `floatwise.models.train`: six quantile models (plus the baselines).
3. `floatwise.models.evaluate`: test forecasts, metrics.json, fairness.json, model_card.md.
4. `floatwise.simulation.backtest`: simulation_default.json with the service-level sweep.
5. `scripts/plot_data_checks.py` and `scripts/build_narratives.py`: plots and cached explanations.

13.3 Run it (about 6 minutes)

```
git clone https://github.com/redwanhasan980/DIU_AI_Hackathon.git
python run.py setup  install Python and web dependencies (~1–2 min)
python run.py all    data, training, evaluation, backtest (~3 min)
python run.py demo   open http://localhost:8000
```

Needs Python 3.11+ and Node.js 20+. No API key or internet needed once installed.

13.4 Quality assurance

Check	Coverage
89 unit and integration tests	Data quality, no leakage, models, engine, allocator, backtest, explanations and LLM guardrails, API contract and errors
Lint	ruff for Python; tsc and eslint for the web app
Browser walk-through (21 checks)	Map, agent, Bangla, approval, DSO limits, live backtest, fairness, decisions, agent card, dark mode

Check	Coverage
CI	GitHub Actions on every push: backend on Linux and Windows, frontend build
Reproducibility	Fresh clone installs and rebuilds the identical model and numbers with a clean git tree

14. Limitations and next steps

14.1 What FloatWise does not fix yet

Limit	Evidence	Response
Some floats are too small for Eid	265–270 of 500 agents still run out on 18–19 March	Escalate as NEEDS_EXTRA_FLOAT (281 agents flagged on 15 March)
Rural and garment areas gain less	−36.9% and −30.3% vs −62.7% to −71.3% elsewhere	Report per segment; fairness review before rollout
A little more cash and travel	+2.3% cash held, +4.1% DSO trips	Service-level dial trades cash for stockouts
Over-confident at spikes	P10–P90 covers 60% of event days (75% overall)	99% service level chosen on validation
Synthetic data, daily granularity	Shaped on public tariffs and network size; no spill-over to neighbouring agents	Shadow mode on real, anonymised data first

14.2 Path to a pilot

1. **Shadow mode (4–6 weeks):** run nightly on one distributor's anonymised daily agent balances and compare with what DSOs actually did.
2. **Pre-Eid pilot:** that distributor's DSOs use the plan; compare stockouts and failed value with control distributors.
3. **Integrate:** DSO and agent apps (“keep Tk X cash tomorrow”), nightly retraining, drift monitoring per area type.
4. **Scale:** one global model for all ~1,54,000 agents with forecasts precomputed overnight, approvals in an audit log.

> **The ask:** one distributor, anonymised daily agent balances and DSO visit logs, and one Eid season, measured against today's rule.

15. Validation of claims

Every claim in this report falls into one of four groups. We kept them separate so that a reader can see exactly what is proven, by whom, and what still needs real data.

Status	Meaning
Verified (external)	Confirmed by an independent public source listed in Appendix C
Verified (our code)	Produced and checked by FloatWise code on synthetic data; reproducible from the repository and covered by tests; not yet tested on real upay data
Assumption	A value we chose because no public source exists; documented in config.yaml and DATA_ASSUMPTIONS.md
Not re-confirmed / Discrepancy	A source was used earlier but could not be re-checked, or the check found a difference

15.1 Facts about Bangladesh, MFS and upay

Claim	Status	Evidence
Cash demand rises sharply around Eid; agents and ATMs are the main cash access when banks close	Verified (external)	Prothom Alo [1]; The Business Standard [5]
Remittances peak before Eid-ul-Fitr	Verified (external)	Prothom Alo [2]; The Business Standard [3]
MFS volumes hit records in Eid months; cash-in/out dominate	Verified (external)	The Daily Star [4]
Salaries and government allowances are paid through MFS	Verified (external)	The Business Standard [6], [10]
Agent liquidity is a known, costly problem in mobile money	Verified (external)	IFC Mobile Money Toolkit [11]
upay has ~1,54,000 agents and ~85 lakh customers (end-2023)	Verified (external)	The Business Standard, 15 Jun 2024 [7]
upay cash-out fee: 1.4% primary, 1.0% salary, 1.0% remittance, 0.7% disbursement wallet	Verified (external)	upay website [8]; The Financial Express, 22 Apr 2021 [9] (1.4%)
Agent commission 0.41% and distributor commission 0.10%	Not re-confirmed	Taken from upay agent information when the data was built; the page no longer shows the rates. Verify with upay. Affects the commission figures and upay's net fee (fee − 0.51%)
Customer cash-out limit Tk 25,000 per transaction	Not re-confirmed	From upay's limits and charges document (UCB); the file could not be downloaded during this check
Eid dates 2024 and 2025 used in the data	Verified (external)	Match the published Bangladesh holiday dates
Eid-ul-Fitr 20 March 2026 and Eid-ul-Adha 27 May 2026 used in the data	Discrepancy	Bangladesh celebrated on 21 March [12] and 28 May 2026 [13], one day later. The dates were set before the moons were sighted. Because the data is synthetic the results are internally consistent, but the 2026 calendar should be corrected (one line in config.yaml) and the pipeline re-run before a real pilot

Claim	Status	Evidence
Eid moves 10–11 days earlier each year	Verified (external)	Follows from the dates above [12]

15.2 Assumptions without a public source

Assumption	Value	Why it is reasonable
Daily volumes per agent and area-type mix	Tk 60,000 – 3,00,000 cash-out per day by archetype	Shapes follow the sources in 2.4; levels are illustrative
Eid, salary and allowance multipliers	e.g. rural cash-out up to ×2.6 before Eid-ul-Fitr	Direction supported by [1]–[6], [10]; size not published
Stockout rates under today's rule (30% pre-Eid, 1.6% normal)	Output of the synthetic ledger	No public agent stockout statistics exist
Today's rule of thumb (top up to 1.5× the 7-day average below 0.6×)	Modelled behaviour	Our model of how agents top up today; not a published upay policy
Customer leaves after a failed cash-out	5%	Chosen by us, not sourced; drives 'customers kept'
Opportunity cost of idle cash	10% a year	Round figure chosen by us, not sourced; drives the idle-cash cost
Allowance (G2P) payment days	Synthetic quarterly schedule	Real schedules vary by programme

15.3 Results

Claim	Status	Evidence
Forecast error 53% lower overall and 64% lower on event days than same day last week	Verified (our code)	artifacts/metrics.json; 89 automated tests; leakage test
Stockouts −47.5%, failed cash-out value −46.7%, upay fee lost −53.7%, cash +2.3%, trips +4.1%	Verified (our code)	artifacts/simulation_default.json; test period never used for tuning
Every area type and rural/urban improve	Verified (our code)	artifacts/fairness.json
About Tk 9.3 crore a month of upay fees recovered at network scale	Derived	Synthetic per-agent result × public agent count [7] × public fees [8] × unconfirmed commission rate. An estimate of the opportunity, not a forecast of upay revenue
Tk 6.4 crore commission, 1.69 lakh stockout-days, 2.7 lakh customers kept per month	Derived	Same scaling; 'customers kept' also depends on the 5% churn assumption
Reproducible build, 89 tests, CI green on Linux and Windows, API responses under 300 ms	Verified (our code)	GitHub Actions; fresh-clone rebuild; browser walk-through

> **What would turn our results into validated results:** a shadow-mode run on one distributor's anonymised daily agent balances, comparing FloatWise's recommendations with what actually happened (Section 14.2). Until then, every result in this report should be read as evidence that the method works on realistic synthetic data.

Appendix A. Glossary

Term	Meaning
Cash-out / cash-in (CO / CI)	Customer withdraws cash from / deposits cash to the agent
E-float	The agent's electronic balance used to serve cash-ins
Stockout	Agent cannot serve a transaction because cash or e-float is exhausted
DSO	Distributor sales officer who visits agents to rebalance cash and e-float
P10 / P50 / P90	Forecast quantiles: 10%, 50% and 90% of outcomes fall below them
WAPE	Weighted absolute percentage error: total absolute error / total actual demand
Service level	Quantile of demand the top-up is sized to cover (0.99 by default)
Censoring	On stockout days the recorded volume understates true demand
G2P	Government-to-person allowance payments

Appendix B. Evidence index

File	Contents
artifacts/metrics.json	Forecast accuracy by slice, coverage, acceptance checks
artifacts/simulation_default.json	Backtest, per-1,000-agent and network-scale value, sweep, segments
artifacts/fairness.json	Per-segment forecast and service checks
artifacts/policy_tuning.json	Validation grid for service level and risk threshold
artifacts/censoring_experiment.json, quantile_experiment.json, allocator_comparison.json	Held-out model and allocator choices
artifacts/model_card.md	Model card
docs/SRS.md, docs/IMPLEMENTATION_PLAN.md, docs/DATA_ASSUMPTIONS.md	Requirements, plan, every assumption with sources

Appendix C. References

All sources were opened and checked on 3 October 2026. Quotations are taken from the articles.

[1] Prothom Alo. “Bank closed for Eid holidays, how to withdraw cash”. 16 March 2026. https://en.prothomalo.com/business/local/kmk5b0q6na

[2] Prothom Alo. “Remittances near $3b mark, set new record before Eid”. 27 March 2025. https://en.prothomalo.com/business/local/3fq2rb78vb

[3] The Business Standard. “Bangladesh receives record $3.29b in remittances in March”. 6 April 2025. https://www.tbsnews.net/economy/bangladesh-receives-record-329-billion-remittances-march-1109181

[4] The Daily Star. “MFS transactions hit record Tk 1.25 lakh crore in April”. 2023. https://www.thedailystar.net/business/economy/news/mfs-transactions-hit-record-tk-125-lakh-crore-april-3352691

[5] The Business Standard. “Cash supply to banks falls short ahead of Eid amid note shortages”. 16 May 2026. https://www.tbsnews.net/economy/cash-supply-banks-falls-short-ahead-eid-amid-note-shortages-1441006

[6] The Business Standard. “MFS reduces RMG workers' payment hassle”. 16 March 2021. https://www.tbsnews.net/economy/mfs-reduces-rmg-workers-payment-hassle-217591

[7] The Business Standard. “upay's accumulated losses cross Tk300cr in 2023”. 15 June 2024. https://www.tbsnews.net/economy/stocks/upays-accumulated-losses-cross-tk300cr-2023-877946

[8] upay. “Cash-out charges by wallet (upaybd.com)”. accessed 3 October 2026. https://www.upaybd.com/services/details/upay-agent

[9] The Financial Express. “'upay' offers lowest cash out charge in mobile money transactions”. 22 April 2021. https://today.thefinancialexpress.com.bd/stock-corporate/upay-offers-lowest-cash-out-charge-in-mobile-money-transactions-1619026261

[10] The Business Standard. “Majority of government allowance recipients opt for Nagad in FY26”. 5 November 2025. https://www.tbsnews.net/economy/corporates/majority-government-allowance-recipients-opt-nagad-fy26-1278111

[11] IFC / World Bank. “Liquidity Management for Mobile Money Providers (IFC Mobile Money Toolkit)”. World Bank Documents. https://documents1.worldbank.org/curated/en/802221501150875893/pdf/117459-WP-Tool-10-5-Liquidity-Management-Series-IFC-mobile-money-toolkit-PUBLIC.pdf

[12] The Daily Star. “Bangladesh to celebrate Eid-ul-Fitr on March 21”. March 2026. https://www.thedailystar.net/news/bangladesh/news/bangladesh-celebrate-eid-ul-fitr-march-21-4132701

[13] Prothom Alo. “Eid-ul-Azha on 28 May as moon sighted”. 18 May 2026. https://en.prothomalo.com/bangladesh/vdn1tkv26i